

# BrickAtlas Hierarchy-3: Complete Supplementary Index

Development Manuscript  
Author metadata intentionally omitted

## 1. Scope and Version

This supplement indexes the current Hierarchy-3 release: 48 layouts, 48 families per layout, and 2,304 questions. It accompanies the machine-readable public inputs, answers, quality records, and original website 3D interface. It replaces the previous manuscript’s appendix as the current entry point. Older Mechanism-1, Hierarchy-1, Hierarchy-2 and retired grid experiments remain in their versioned data directories and Git history; their scores are not pooled into this release.

## 2. Reading the Contracts

Each task has a source ID, difficulty stratum, task layer, family, capability, output format, evidence class, and public input. Single choices return `choiceId`; set-valued choices return `choiceIds`; operations return `actionIds`; schedules return a job-to-start-time map; policies return a query and observation-to-action map. Choice values are shuffled deterministically per task. Empty feasible sets are accepted when appropriate.

The full companion `QUESTION_BANK.zh-CN.md` includes every question, every option, every input, and every reference answer. The following family table describes all contracts, and the per-object pages index all 2,304 task instances with four actual website renders per object. No page implies an independent human annotation.

## 3. Atomic Operations

Identification maps an image-highlighted service module to its ID. Counting groups visible parts by module. Color and position questions read explicit attributes and world bounding-box centers. Joint type, parents, roots, and incident-degree questions inspect the declared attachment graph.

Recoloring changes exactly one part’s color. Addition chooses an exact missing part descriptor. Removal identifies leaf modules. Replacement maximizes stiffness under a cost limit. Translation inverts an observed offset; rotation selects a shortest signed yaw correction. Next-module questions test predecessor completion. Inventory checks conserve piece count; boundary questions identify every in-

cident joint; no-operation questions reward preserving an already-correct state. Local-frame, orthographic projection, relative-order and joint freedom tasks extend the atomic coverage.

These are controlled contracts, and several public inputs disclose the exact attributes needed for a solution. They are not all image-only perception tasks. One must report the information condition alongside accuracy and run ablations before assigning a visual interpretation.

## 4. Meta and Physical Tasks

Prefix tasks detect the first violated predecessor. Accessibility reads actual continuous cuboid shape-cast records. Counterfactual tasks remove a specified node and recompute undirected reachability to anchored roots; the explicitly removed node is excluded from the answer. Dynamics asks for the largest sampled translational displacement from eleven actual Rapier measurements.

Kinematic questions use declared joint axes and limits, not empirical actuation success. Diagnosis matches constraint type and observed endpoints, accepting all matching interfaces. Inspection optimizes entropy reduction per unit query cost. Abstention checks agreement of actions across possible worlds. Posterior inference conditions a finite uniform prior on an observation. Pareto enumeration retains all non-dominated cost/mass/stiffness alternatives. New families cover clearance margins, cross-product moments, weighted posteriors, expected losses, and sampled threshold exceedance.

## 5. Action State Machine

The initial fact set, action catalog, preconditions, forbidden facts, add/delete effects, costs, budget, and goals are public. Assembly/disassembly changes module presence. A service intervention requires support before access, access before removal or modification, replacement before verification, closure before support release. Guarded repair and rollback expose invalid shortcuts, while shared-tool repair enforces exclusive access through a resource fact. Actions on independent modules may be interleaved. The evaluator never sorts or repairs a submitted program.

For every action, the replay records accepted status, first error, current facts, visible modules, color overrides, and cumulative cost. Reference and submitted model JSON use the same interpreter in the original webpage. A visible explosion is a display transformation only. Neither symbolic acceptance nor rendered motion is claimed to certify clutch forces or robot insertion.

## 6. Integration

Multi-fault programs combine several targets with one budget and final goal. Scheduling checks integer starts, predecessor finishes, resource exclusion, and makespan. Policies must map all observations to actions correct in every compatible world; success on one sampled branch is insufficient. Unknown actions, repeated completed actions, omitted terminal actions, conflicts, and missing policy branches are tested with negative controls.

## 7. Simulator and Quality Boundaries

Rapier 0.20.0 runs at 120 Hz and 64 solver iterations. Modules are compound stud-inclusive cuboid bodies. Bases are fixed. ForceBased holding motors use stiffness 60000, damping 1500, and maximum force 20000 in scene units. Three access casts use a 0.18-unit half-extent probe and exclude the service module. The static query world is independent of settling.

After 120 nominal steps, an impulse (10, 1.8, 3.5) is applied one unit above the target geometric representative point. The same attached point supplies the displacement measurements. The next 120 steps produce eleven samples. Nominal state is evaluated before the impulse. All 48 layouts satisfy representative-point displacement below 0.35, angular displacement below 0.15 rad and non-spring joint residual below 0.02. Initial anchor pairs coincide. No module pair has oriented-cuboid penetration greater than 0.005. Direct joint interfaces participate in the collision gate and retain contacts; module interiors are compound rigid bodies. Revolute limits are explicitly enabled on instantiated joints and read back. A driven motor is stopped at its declared bound, and changing only the local coordinate origin preserves the sampled physical response. Geometry, anchors, some layouts, moorings and motor parameters changed together, so comparisons with Hierarchy-2 require the new protocol label. This does not certify commercial connector fit or arbitrary robot motion.

## 8. Corrections and Historical Scoring

Hierarchy-2 had eleven next-module questions with semantically equal set options in different orders. The corrected generator normalizes sets only; vectors and programs remain ordered. Historical affected choice IDs are recorded

in `historical-ambiguities.json`; either equivalent answer is accepted under a separately labeled semantic rescoring. Historical raw scores are retained.

The Mechanism-1 conditional-entropy sign was corrected. An uninformative four-world query now has information gain zero rather than four; the six historical optimal query labels did not change. All primitive dimensions now agree with the renderer and stud-inclusive envelopes. Paused replay selection and isolation cannot resurrect absent modules. Six old constant-answer families now vary actual parameters, and all 48 semantic answer distributions are published.

## 9. Pilot and Reproduction

The GPT-4.1 mini pilot contains 16 fixed questions, one per level/layer cell. Families are local-frame, expected-loss, guarded-repair, and resource-repair. 7 of 16 responses succeed at a total cost of \$0.013621. Each cell has one response, so no stable accuracy estimate or model ranking follows. Input hashes, generation IDs, provider receipts, and raw predictions are archived separately from the reference answers.

The 2,304 oracle checks validate the evaluator; 384 public-precondition solutions validate action execution. A constant-choice/empty-answer control passes 430 cases. These controls do not establish task novelty or human difficulty. The figures use original-site canvas captures, and byte-identical model records are shipped to the website and offline release.

From the repository root:

```
npm --prefix benchmark run h3:release
npm run start:local
npm --prefix benchmark run h3:render
npm --prefix benchmark run h3:docs
npm --prefix benchmark test
npm run build
```

Model inference is a separate paid command and is not required for offline score replay. The release manifest binds data and generator hashes. Public-input files exclude the answer; the review website intentionally includes a reference-answer control.

## 10. All 48 Task Families

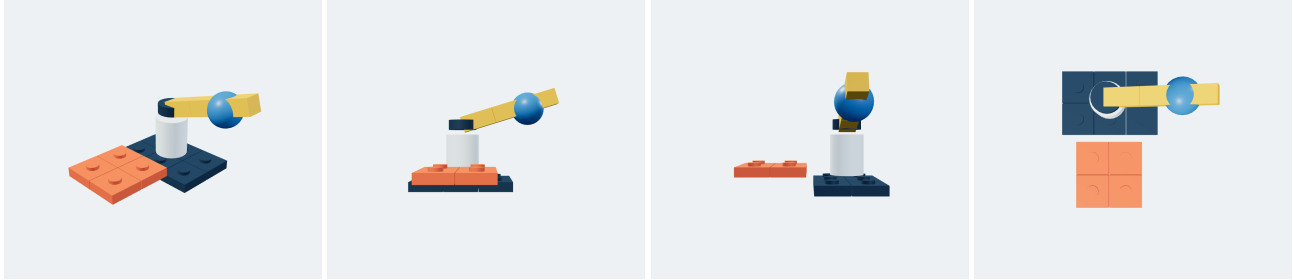
| Family             | Layer         | Output          | Evidence          |
|--------------------|---------------|-----------------|-------------------|
| identify           | atomic        | single-choice   | model-state       |
| count              | atomic        | single-choice   | model-state       |
| color              | atomic        | single-choice   | model-state       |
| position           | atomic        | single-choice   | model-state       |
| joint-type         | atomic        | single-choice   | model-state       |
| parent             | atomic        | single-choice   | model-state       |
| anchor             | atomic        | single-choice   | model-state       |
| degree             | atomic        | single-choice   | model-state       |
| recolor            | atomic        | single-choice   | model-state       |
| add                | atomic        | single-choice   | model-state       |
| remove             | atomic        | single-choice   | model-state       |
| replace            | atomic        | single-choice   | model-state       |
| translate          | atomic        | single-choice   | model-state       |
| rotate             | atomic        | single-choice   | model-state       |
| next-module        | atomic        | single-choice   | model-state       |
| inventory          | atomic        | single-choice   | model-state       |
| boundary           | atomic        | single-choice   | model-state       |
| no-op              | atomic        | single-choice   | model-state       |
| prefix             | metacognitive | single-choice   | support-graph     |
| access             | metacognitive | multiple-choice | Rapier            |
| counterfactual     | metacognitive | single-choice   | support-graph     |
| dynamic            | metacognitive | single-choice   | Rapier            |
| kinematic          | metacognitive | multiple-choice | model-state       |
| diagnosis          | metacognitive | multiple-choice | model-state       |
| information-gain   | metacognitive | multiple-choice | finite-world      |
| abstention         | metacognitive | single-choice   | finite-world      |
| posterior          | metacognitive | single-choice   | finite-world      |
| pareto             | metacognitive | multiple-choice | model-state       |
| assembly           | procedural    | actions         | state-machine     |
| disassembly        | procedural    | actions         | state-machine     |
| service-repair     | procedural    | actions         | state-machine     |
| compound-edit      | procedural    | actions         | state-machine     |
| multi-fault        | integrative   | actions         | state-machine     |
| scheduling         | integrative   | schedule        | resource-schedule |
| policy             | integrative   | policy          | finite-world      |
| local-frame        | atomic        | single-choice   | model-state       |
| projection         | atomic        | single-choice   | model-state       |
| relative-order     | atomic        | single-choice   | model-state       |
| joint-axis         | atomic        | single-choice   | model-state       |
| clearance-margin   | metacognitive | single-choice   | model-state       |
| load-moment        | metacognitive | single-choice   | model-state       |
| weighted-posterior | metacognitive | single-choice   | finite-world      |
| expected-loss      | metacognitive | single-choice   | finite-world      |
| trace-threshold    | metacognitive | single-choice   | Rapier            |
| guarded-repair     | procedural    | actions         | state-machine     |
| rollback           | procedural    | actions         | state-machine     |
| resource-repair    | integrative   | actions         | state-machine     |
| budget-policy      | integrative   | policy          | finite-world      |

Each row appears on twelve source layouts at each of four levels. Rows sharing an object remain correlated; source-level reporting and family-grouped confirmation are required. The per-object pages below are the complete release index. A model's render count is not a source count, and shared part counts do not imply equivalent task difficulty.

## 11. Complete Object and Question Index

### 11.1. 1. Camera Gimbal (D1)

16 visible parts, 4 modules, 3 joints. Nominal representative-point displacement: 0.0006; angular displacement: 0.0000 rad. No inter-module penetration above 0.005, including directly joined modules.

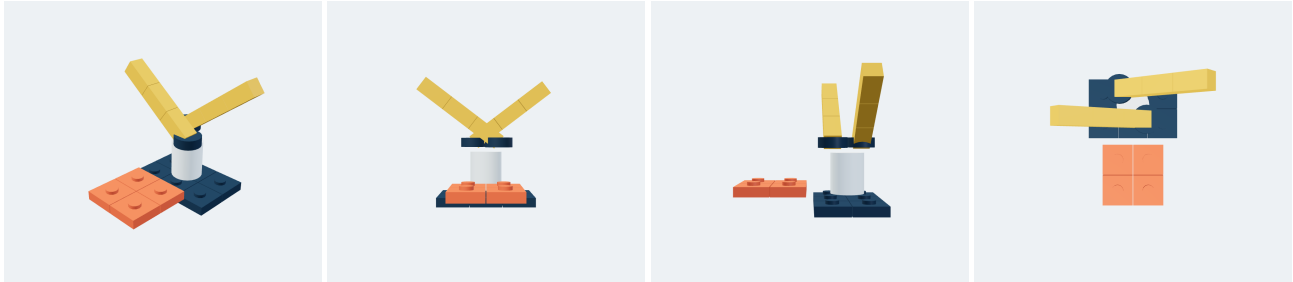


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

## 11.2. 2. Folding Gripper (D1)

19 visible parts, 5 modules, 4 joints. Nominal representative-point displacement: 0.0005; angular displacement: 0.0000 rad. No inter-module penetration above 0.005, including directly joined modules.

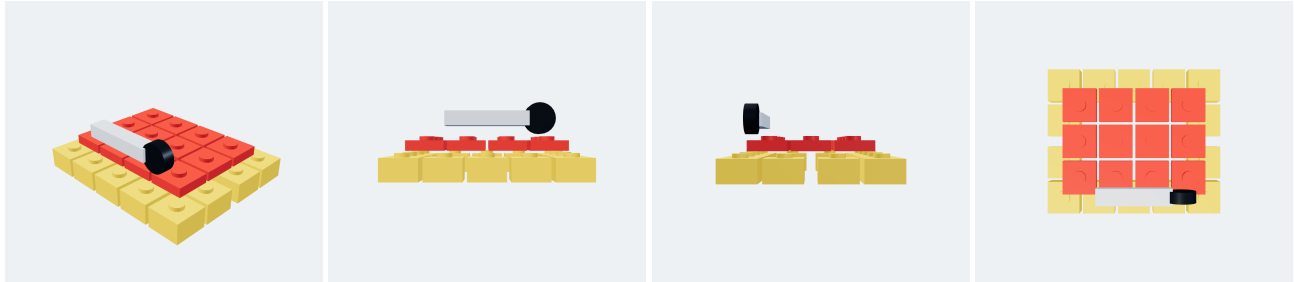


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.3. 3. Hinged Safety Hatch (D1)

28 visible parts, 3 modules, 2 joints. Nominal representative-point displacement: 0.0000; angular displacement: 0.0000 rad. No inter-module penetration above 0.005, including directly joined modules.



All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

#### 11.4. 4. Indexed Toggle Latch (D1)

25 visible parts, 6 modules, 5 joints. Nominal representative-point displacement: 0.0000; angular displacement: 0.0000 rad. No inter-module penetration above 0.005, including directly joined modules.

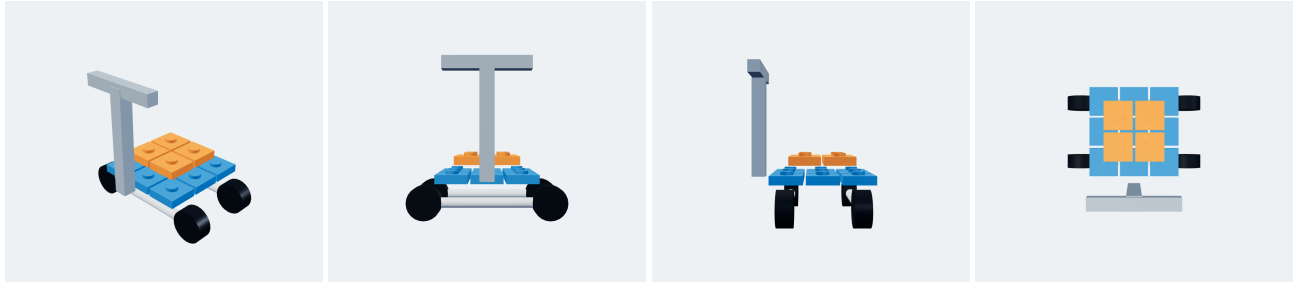


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.5. 5. Maintenance Trolley (D1)

21 visible parts, 5 modules, 4 joints. Nominal representative-point displacement: 0.0000; angular displacement: 0.0000 rad. No inter-module penetration above 0.005, including directly joined modules.

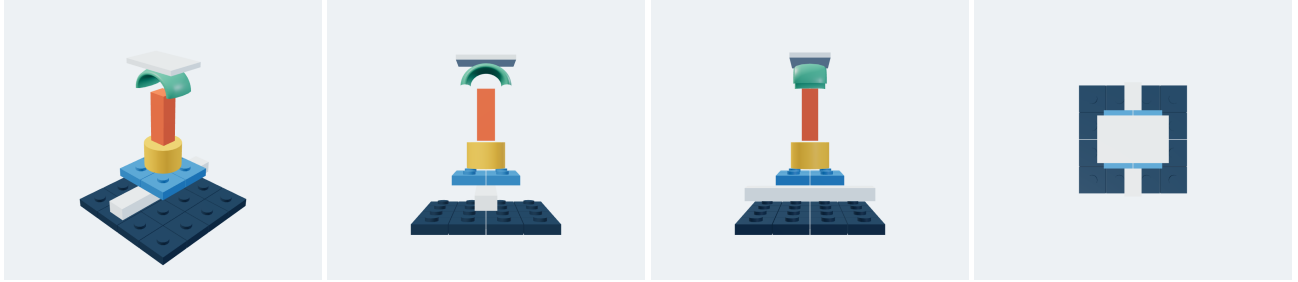


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.6. 6. Orthogonal Probe Stage (D1)

25 visible parts, 6 modules, 5 joints. Nominal representative-point displacement: 0.0000; angular displacement: 0.0000 rad. No inter-module penetration above 0.005, including directly joined modules.

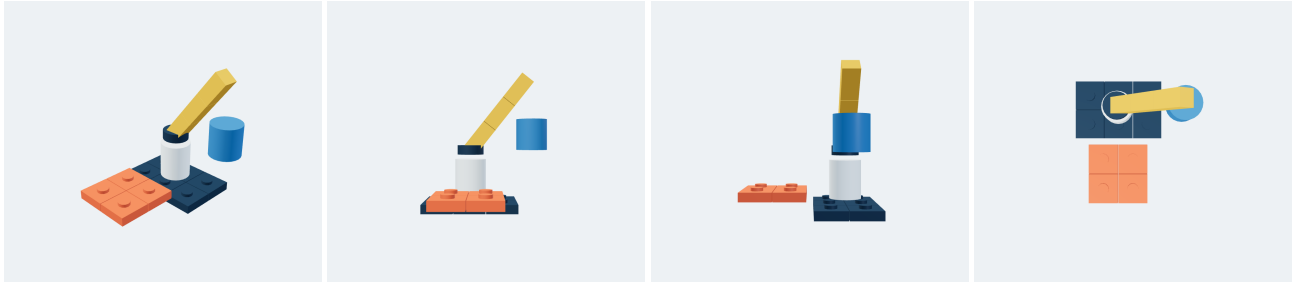


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.7. 7. Ratchet Hoist (D1)

16 visible parts, 4 modules, 3 joints. Nominal representative-point displacement: 0.0006; angular displacement: 0.0000 rad. No inter-module penetration above 0.005, including directly joined modules.

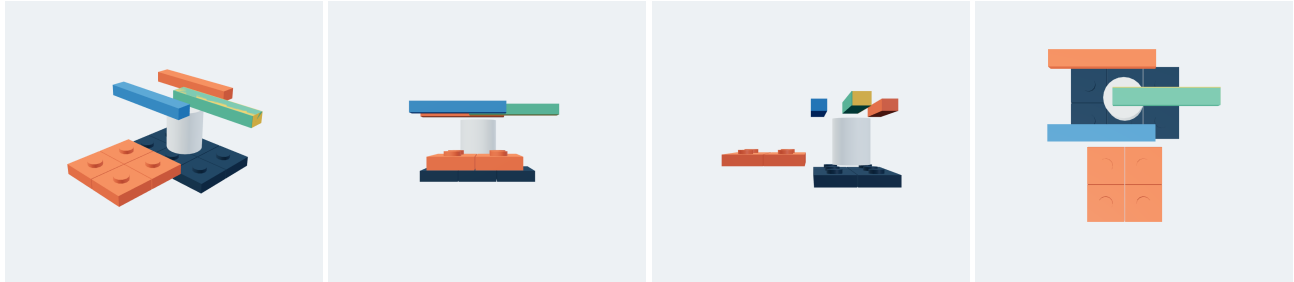


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.8. 8. Sample Carousel (D1)

15 visible parts, 4 modules, 3 joints. Nominal representative-point displacement: 0.0000; angular displacement: 0.0000 rad. No inter-module penetration above 0.005, including directly joined modules.

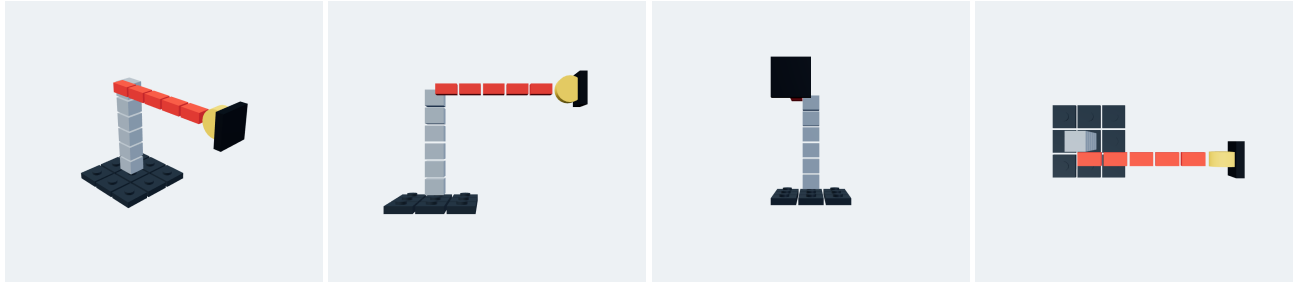


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.9. 9. Signal Switch Stand (D1)

22 visible parts, 4 modules, 3 joints. Nominal representative-point displacement: 0.0035; angular displacement: 0.0000 rad. No inter-module penetration above 0.005, including directly joined modules.

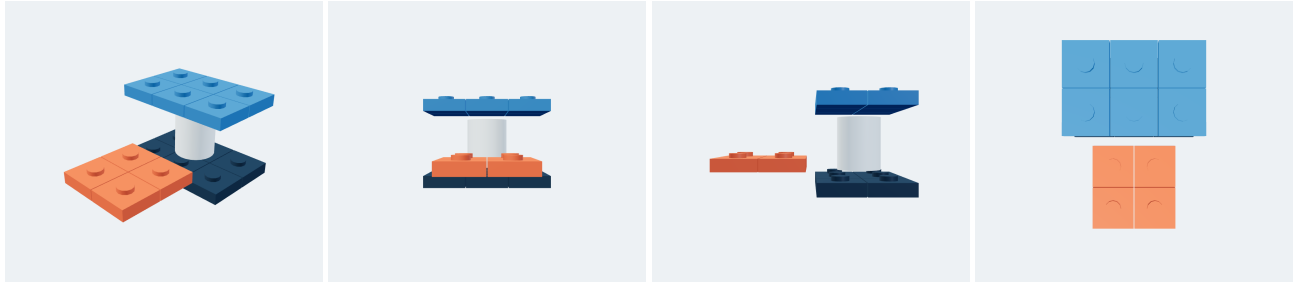


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.10. 10. Rail Tool Drawer (D1)

17 visible parts, 4 modules, 3 joints. Nominal representative-point displacement: 0.0000; angular displacement: 0.0000 rad. No inter-module penetration above 0.005, including directly joined modules.

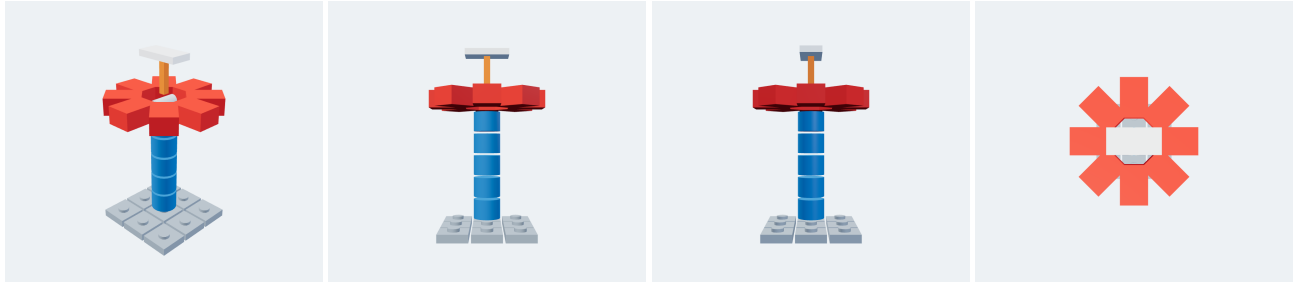


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.11. 11. Valve Control Stand (D1)

25 visible parts, 3 modules, 2 joints. Nominal representative-point displacement: 0.0000; angular displacement: 0.0000 rad. No inter-module penetration above 0.005, including directly joined modules.

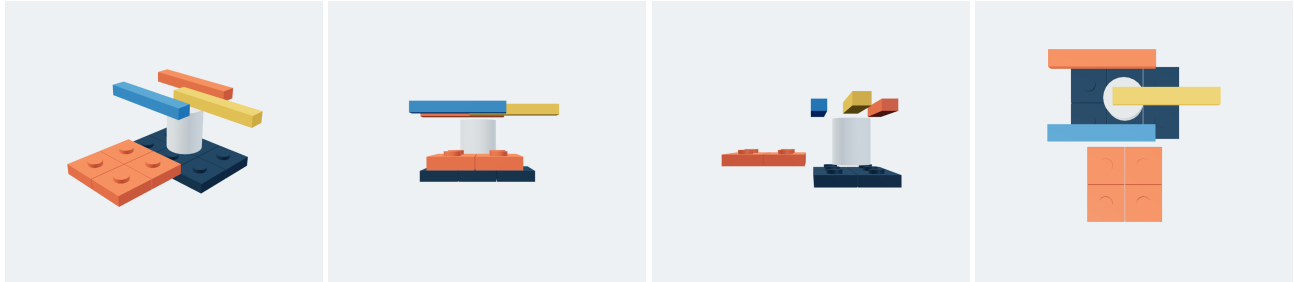


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.12. 12. Wind Vane (D1)

14 visible parts, 4 modules, 3 joints. Nominal representative-point displacement: 0.0000; angular displacement: 0.0000 rad. No inter-module penetration above 0.005, including directly joined modules.



All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.13. 13. Articulated Camera Dolly (D2)

54 visible parts, 11 modules, 10 joints. Nominal representative-point displacement: 0.0000; angular displacement: 0.0000 rad. No inter-module penetration above 0.005, including directly joined modules.

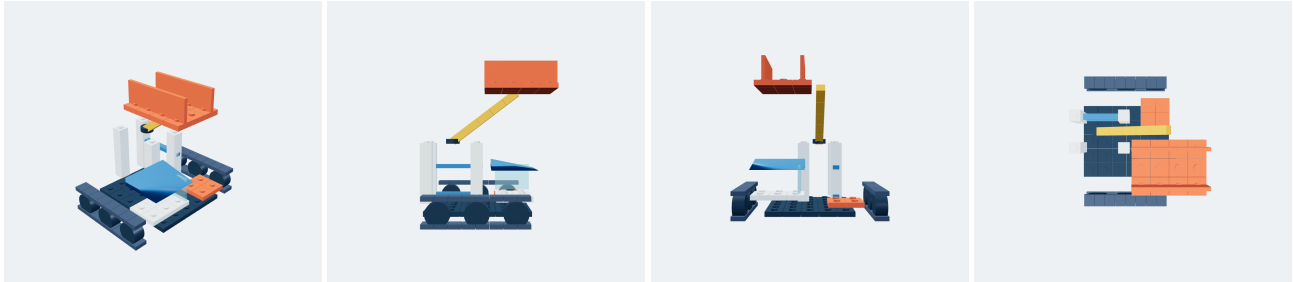


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.14. 14. Articulated Loader (D2)

126 visible parts, 8 modules, 7 joints. Nominal representative-point displacement: 0.0100; angular displacement: 0.0017 rad. No inter-module penetration above 0.005, including directly joined modules.

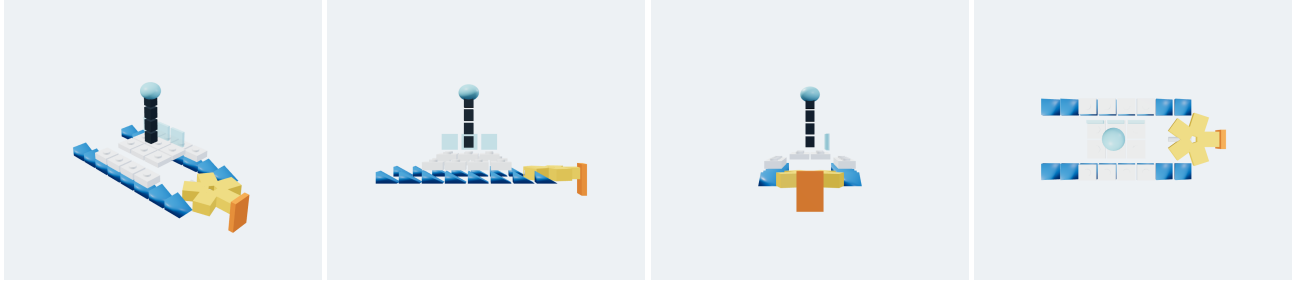


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.15. 15. Canal Inspection Skiff (D2)

45 visible parts, 5 modules, 4 joints. Nominal representative-point displacement: 0.0000; angular displacement: 0.0000 rad. No inter-module penetration above 0.005, including directly joined modules.

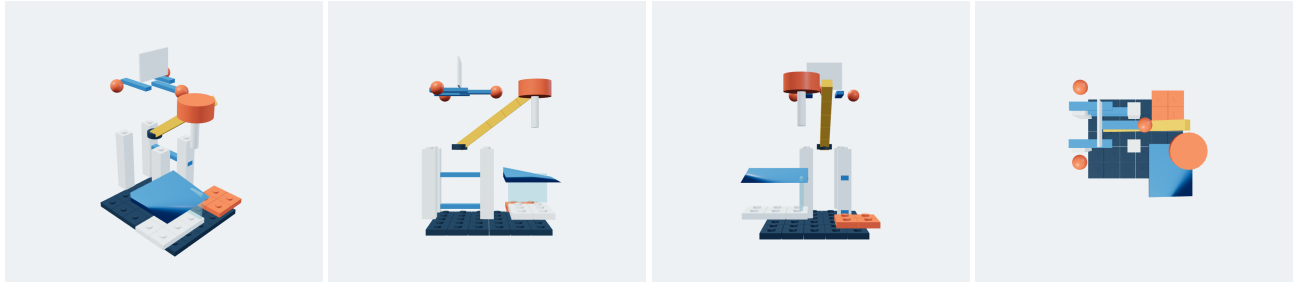


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.16. 16. Folding Weather Mast (D2)

83 visible parts, 7 modules, 6 joints. Nominal representative-point displacement: 0.0084; angular displacement: 0.0017 rad. No inter-module penetration above 0.005, including directly joined modules.



All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.17. 17. Parallel Pallet Handler (D2)

54 visible parts, 11 modules, 10 joints. Nominal representative-point displacement: 0.0000; angular displacement: 0.0000 rad. No inter-module penetration above 0.005, including directly joined modules.

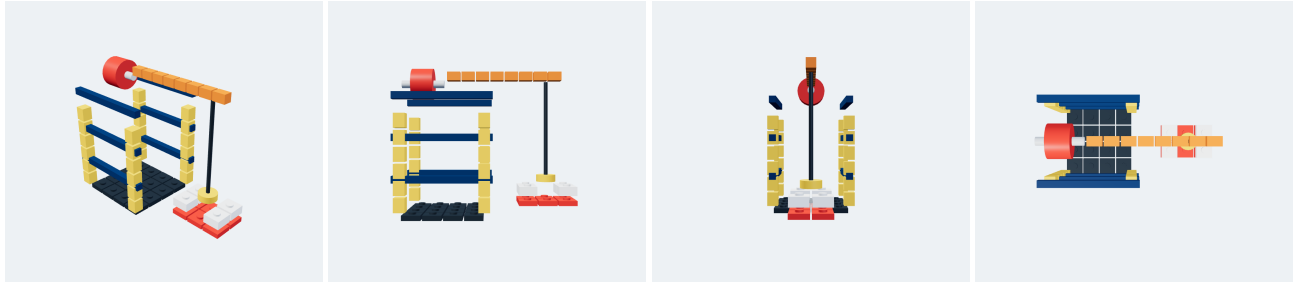


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.18. 18. Rescue Winch Tower (D2)

68 visible parts, 6 modules, 5 joints. Nominal representative-point displacement: 0.0530; angular displacement: 0.0065 rad.  
No inter-module penetration above 0.005, including directly joined modules.

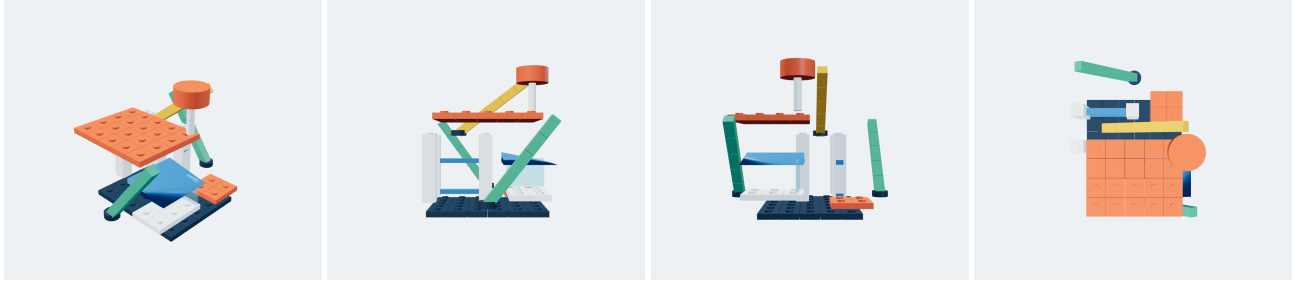


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.19. 19. Scissor Service Lift (D2)

108 visible parts, 9 modules, 8 joints. Nominal representative-point displacement: 0.0084; angular displacement: 0.0017 rad. No inter-module penetration above 0.005, including directly joined modules.

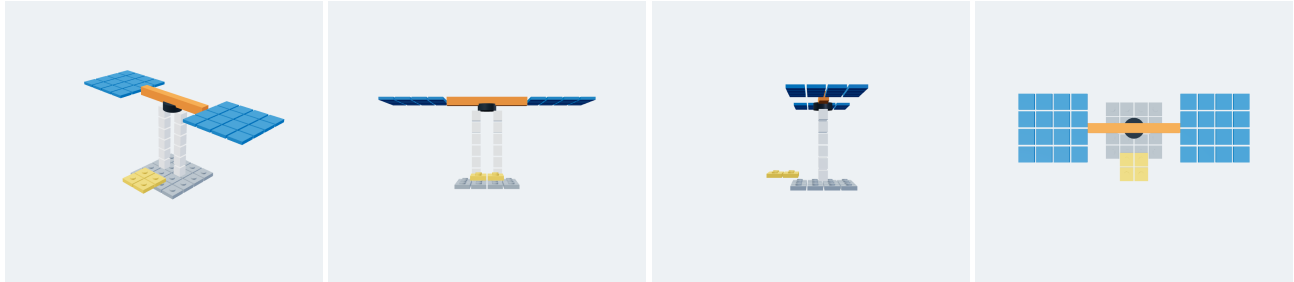


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

## 11.20. 20. Solar Tracker Array (D2)

66 visible parts, 6 modules, 5 joints. Nominal representative-point displacement: 0.0000; angular displacement: 0.0000 rad. No inter-module penetration above 0.005, including directly joined modules.

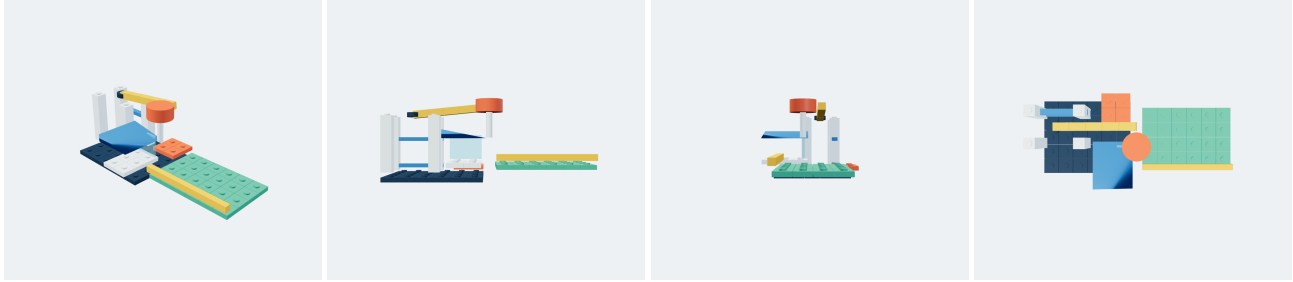


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.21. 21. Swing Loading Dock (D2)

101 visible parts, 7 modules, 6 joints. Nominal representative-point displacement: 0.0115; angular displacement: 0.0022 rad. No inter-module penetration above 0.005, including directly joined modules.

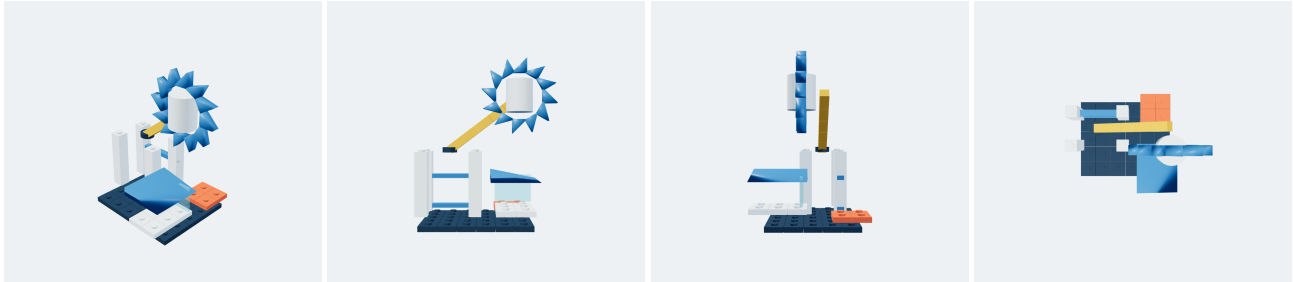


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

## 11.22. 22. Telescopic Optics Cart (D2)

87 visible parts, 6 modules, 5 joints. Nominal representative-point displacement: 0.0093; angular displacement: 0.0017 rad. No inter-module penetration above 0.005, including directly joined modules.

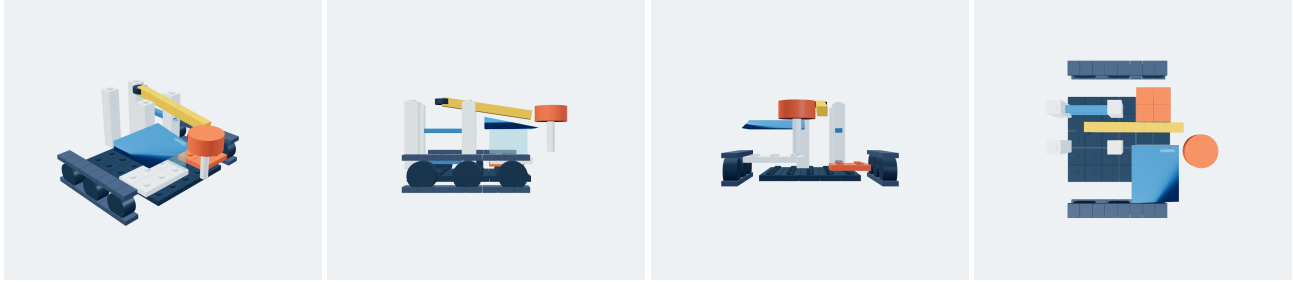


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.23. 23. Tracked Core Drill (D2)

114 visible parts, 8 modules, 7 joints. Nominal representative-point displacement: 0.0159; angular displacement: 0.0026 rad. No inter-module penetration above 0.005, including directly joined modules.

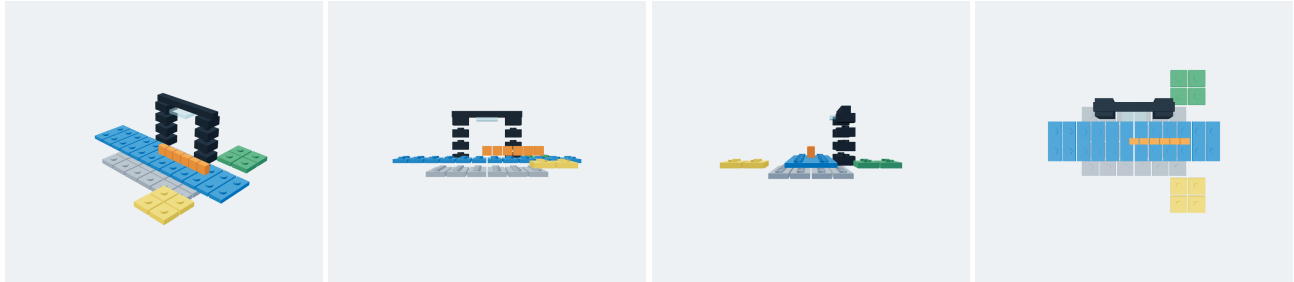


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.24. 24. Warehouse Sorter (D2)

60 visible parts, 6 modules, 5 joints. Nominal representative-point displacement: 0.0000; angular displacement: 0.0000 rad. No inter-module penetration above 0.005, including directly joined modules.

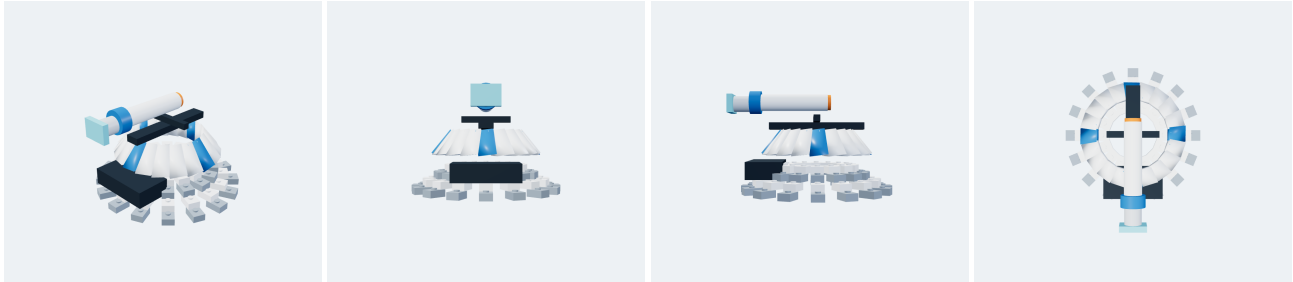


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.25. 25. Adaptive Radio Observatory (D3)

92 visible parts, 4 modules, 3 joints. Nominal representative-point displacement: 0.0129; angular displacement: 0.0022 rad. No inter-module penetration above 0.005, including directly joined modules.



All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.26. 26. Bascule Canal Gate (D3)

142 visible parts, 6 modules, 7 joints. Nominal representative-point displacement: 0.0019; angular displacement: 0.0000 rad. No inter-module penetration above 0.005, including directly joined modules.

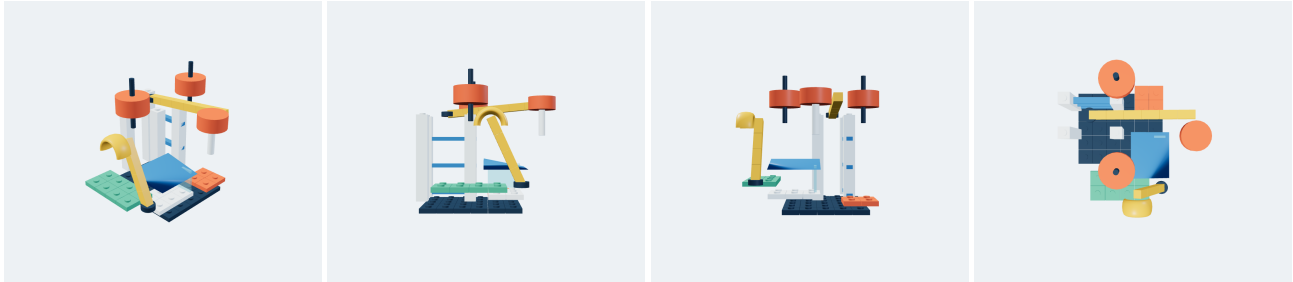


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.27. 27. Cable Inspection Robot (D3)

108 visible parts, 11 modules, 10 joints. Nominal representative-point displacement: 0.0167; angular displacement: 0.0028 rad. No inter-module penetration above 0.005, including directly joined modules.

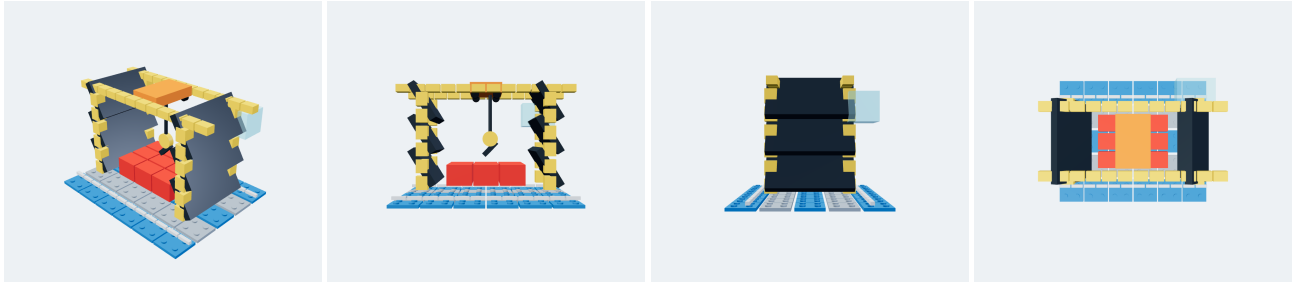


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.28. 28. Harbor Container Crane (D3)

126 visible parts, 5 modules, 5 joints. Nominal representative-point displacement: 0.0014; angular displacement: 0.0000 rad. No inter-module penetration above 0.005, including directly joined modules.

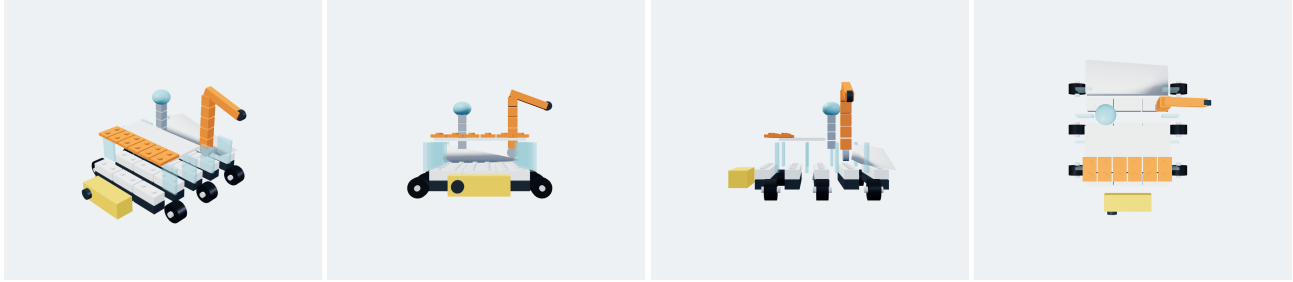


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.29. 29. Orbital Service Rover (D3)

73 visible parts, 10 modules, 9 joints. Nominal representative-point displacement: 0.0004; angular displacement: 0.0000 rad. No inter-module penetration above 0.005, including directly joined modules.

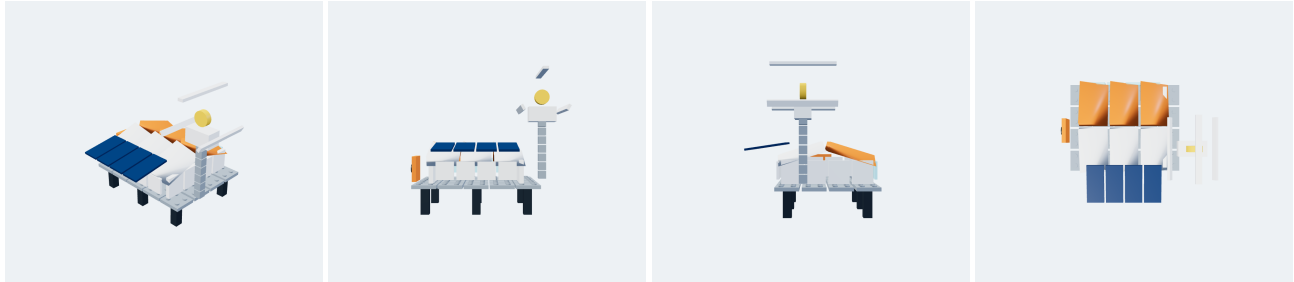


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.30. 30. Polar Research Station (D3)

66 visible parts, 5 modules, 4 joints. Nominal representative-point displacement: 0.0000; angular displacement: 0.0000 rad. No inter-module penetration above 0.005, including directly joined modules.

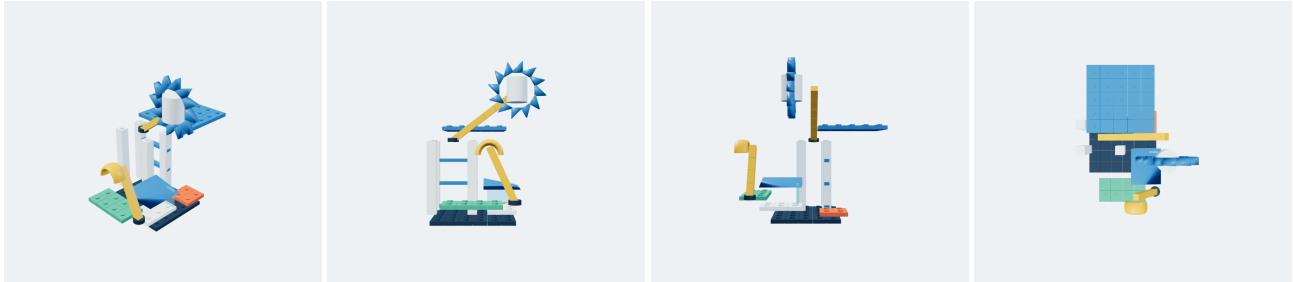


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.31. 31. Retractable Array Radar (D3)

140 visible parts, 10 modules, 9 joints. Nominal representative-point displacement: 0.0163; angular displacement: 0.0028 rad. No inter-module penetration above 0.005, including directly joined modules.

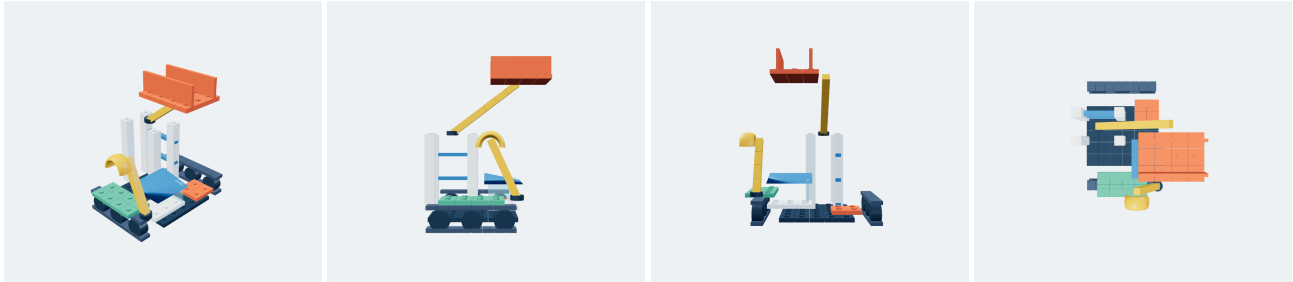


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.32. 32. Rotary Bucket Excavator (D3)

154 visible parts, 11 modules, 10 joints. Nominal representative-point displacement: 0.0144; angular displacement: 0.0022 rad. No inter-module penetration above 0.005, including directly joined modules.



All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.33. 33. Segmented Petal Observatory (D3)

196 visible parts, 43 modules, 42 joints. Nominal representative-point displacement: 0.0000; angular displacement: 0.0000 rad. No inter-module penetration above 0.005, including directly joined modules.

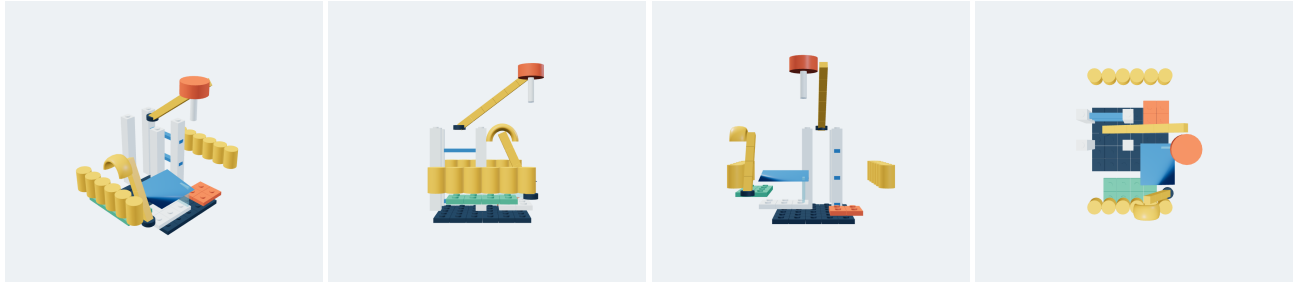


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.34. 34. Subsea Manipulator (D3)

116 visible parts, 11 modules, 10 joints. Nominal representative-point displacement: 0.0124; angular displacement: 0.0022 rad. No inter-module penetration above 0.005, including directly joined modules.



All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.35. 35. Tri-arm Sample Handler (D3)

130 visible parts, 22 modules, 21 joints. Nominal representative-point displacement: 0.0000; angular displacement: 0.0000 rad. No inter-module penetration above 0.005, including directly joined modules.

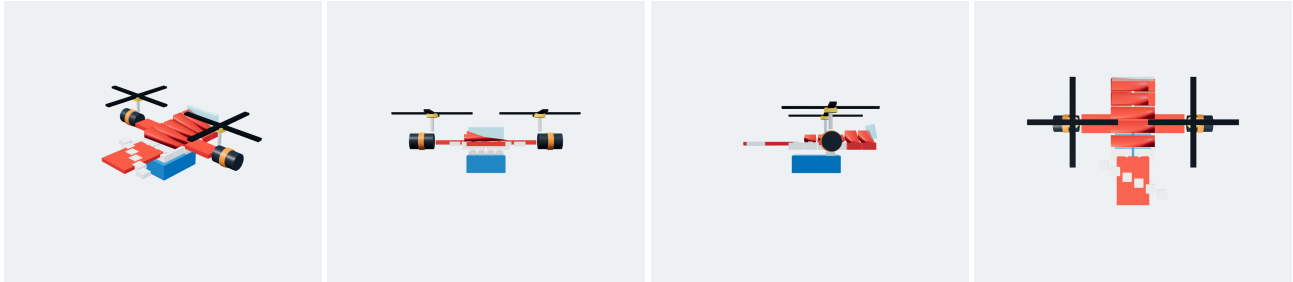


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.36. 36. Wildfire Tiltrotor (D3)

43 visible parts, 6 modules, 5 joints. Nominal representative-point displacement: 0.0004; angular displacement: 0.0000 rad. No inter-module penetration above 0.005, including directly joined modules.

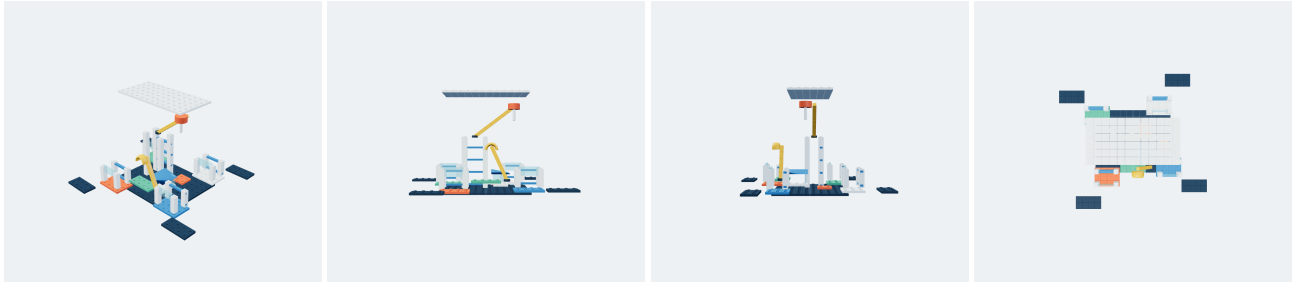


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.37. 37. Aircraft Service Hangar (D4)

437 visible parts, 22 modules, 21 joints. Nominal representative-point displacement: 0.0173; angular displacement: 0.0026 rad. No inter-module penetration above 0.005, including directly joined modules.



All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.38. 38. Automated Cargo Terminal (D4)

231 visible parts, 13 modules, 12 joints. Nominal representative-point displacement: 0.0006; angular displacement: 0.0000 rad. No inter-module penetration above 0.005, including directly joined modules.

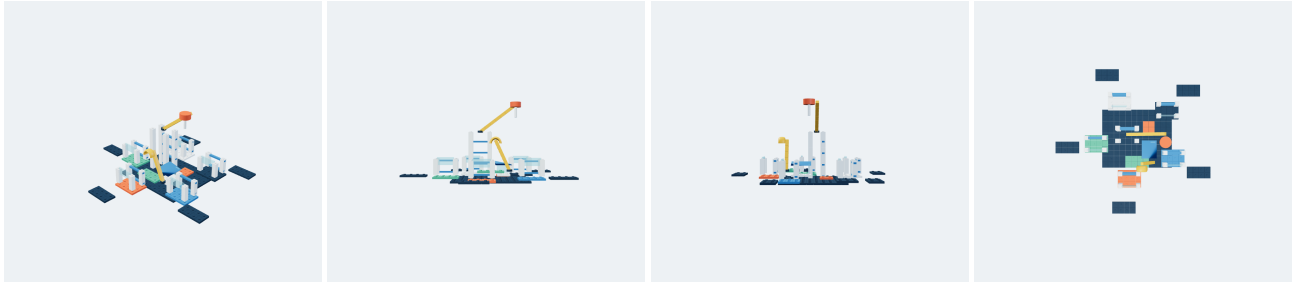


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.39. 39. Automated Recycling Plant (D4)

404 visible parts, 24 modules, 23 joints. Nominal representative-point displacement: 0.0173; angular displacement: 0.0026 rad. No inter-module penetration above 0.005, including directly joined modules.

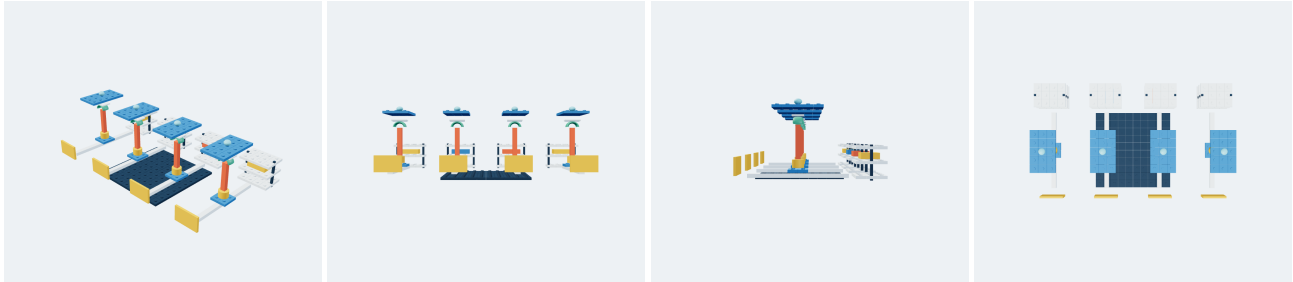


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

#### 11.40. 40. Distributed Battery Exchange (D4)

360 visible parts, 41 modules, 40 joints. Nominal representative-point displacement: 0.0000; angular displacement: 0.0000 rad. No inter-module penetration above 0.005, including directly joined modules.

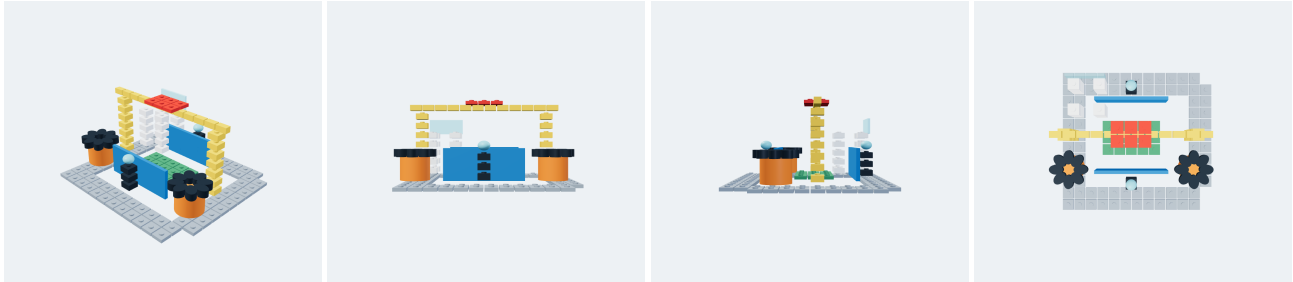


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.41. 41. Flood Response Lock (D4)

214 visible parts, 11 modules, 12 joints. Nominal representative-point displacement: 0.0090; angular displacement: 0.0182 rad. No inter-module penetration above 0.005, including directly joined modules.

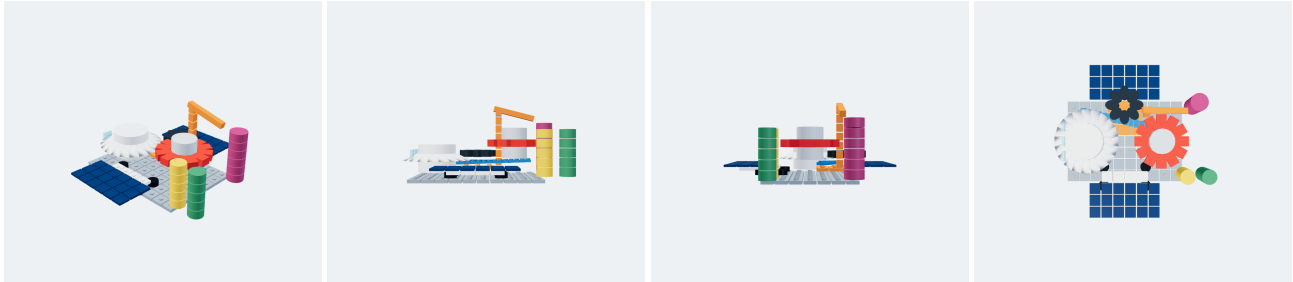


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

## 11.42. 42. Lunar Sample Refinery (D4)

218 visible parts, 12 modules, 11 joints. Nominal representative-point displacement: 0.0002; angular displacement: 0.0076 rad. No inter-module penetration above 0.005, including directly joined modules.

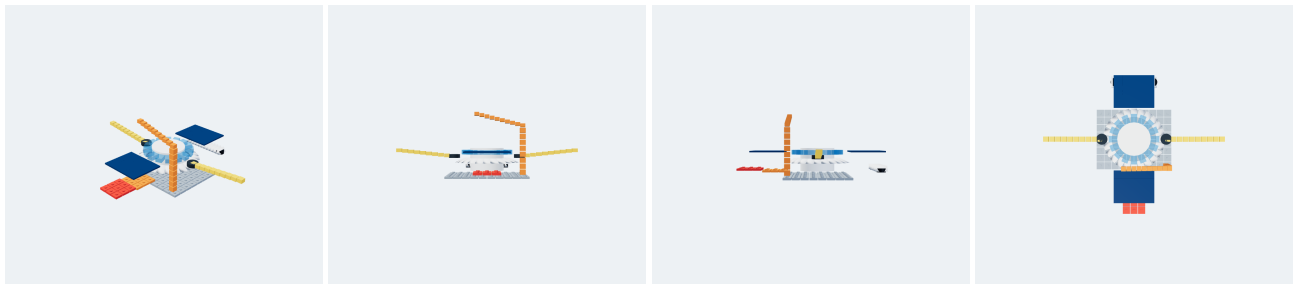


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.43. 43. Orbital Docking Yard (D4)

233 visible parts, 11 modules, 10 joints. Nominal representative-point displacement: 0.0051; angular displacement: 0.0020 rad. No inter-module penetration above 0.005, including directly joined modules.

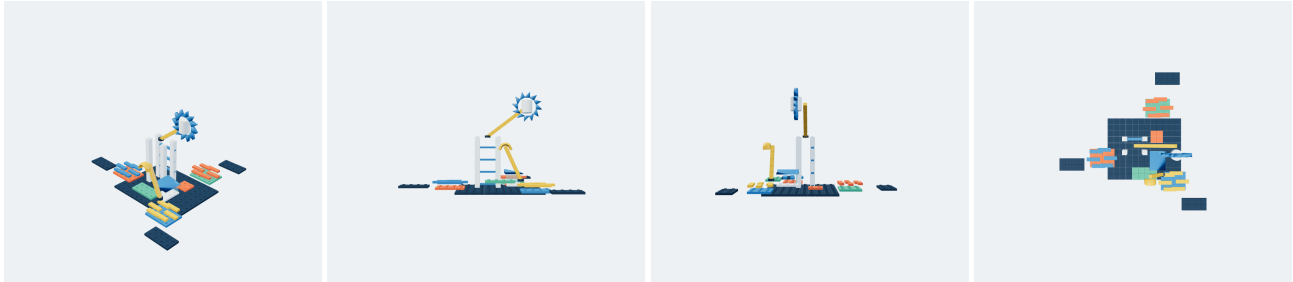


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

#### 11.44. 44. Orbital Telescope Yard (D4)

298 visible parts, 18 modules, 17 joints. Nominal representative-point displacement: 0.0185; angular displacement: 0.0026 rad. No inter-module penetration above 0.005, including directly joined modules.

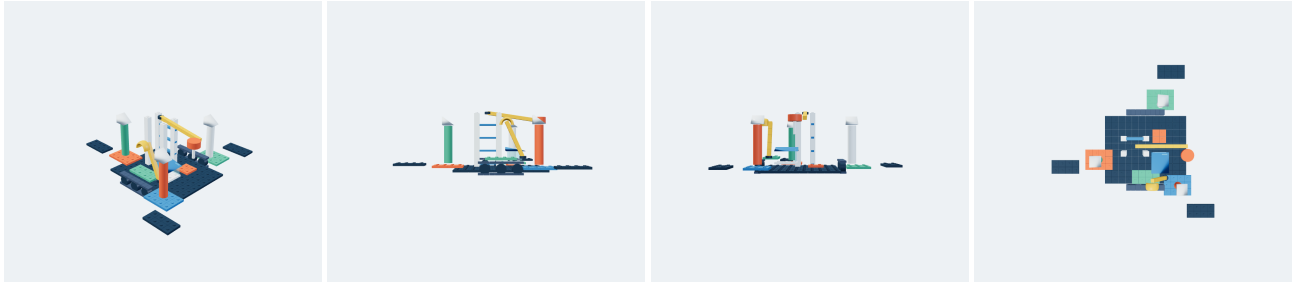


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.45. 45. Polar Drilling Complex (D4)

331 visible parts, 20 modules, 19 joints. Nominal representative-point displacement: 0.0239; angular displacement: 0.0032 rad. No inter-module penetration above 0.005, including directly joined modules.

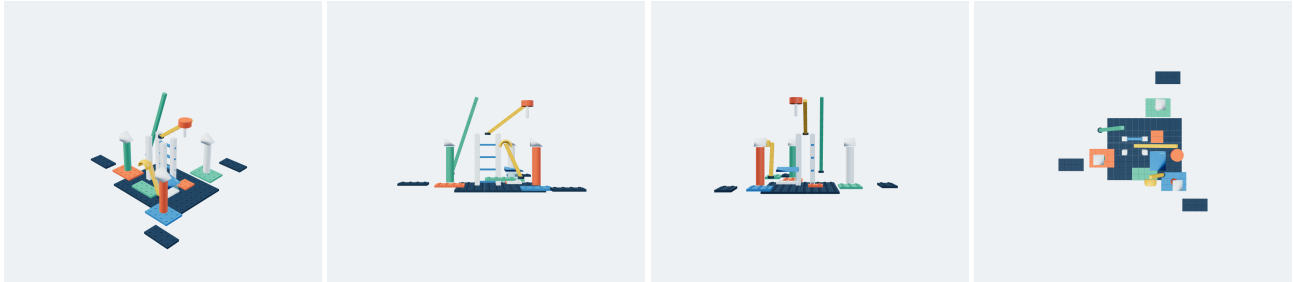


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.46. 46. Rescue Launch Complex (D4)

306 visible parts, 19 modules, 18 joints. Nominal representative-point displacement: 0.0173; angular displacement: 0.0026 rad. No inter-module penetration above 0.005, including directly joined modules.

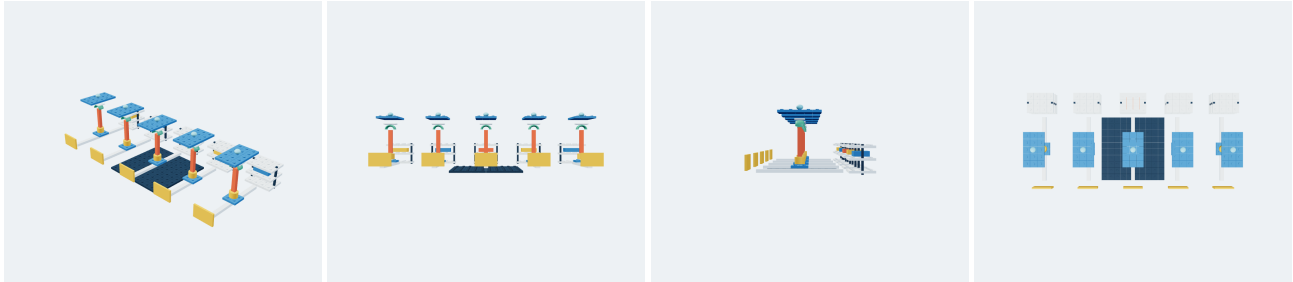


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.47. 47. Segmented Antenna Yard (D4)

425 visible parts, 51 modules, 50 joints. Nominal representative-point displacement: 0.0000; angular displacement: 0.0000 rad. No inter-module penetration above 0.005, including directly joined modules.

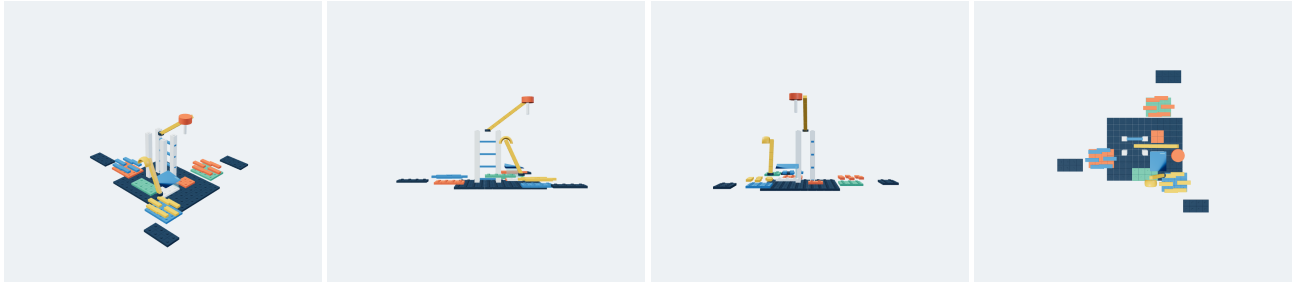


All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |

### 11.48. 48. Tidal Energy Station (D4)

287 visible parts, 18 modules, 17 joints. Nominal representative-point displacement: 0.0173; angular displacement: 0.0026 rad. No inter-module penetration above 0.005, including directly joined modules.



All 48 tasks share this source; complete inputs, choices and answers are in the companion question bank.

| No. | Family             | Layer         | Output          | Evidence          |
|-----|--------------------|---------------|-----------------|-------------------|
| 1   | identify           | atomic        | single-choice   | model-state       |
| 2   | count              | atomic        | single-choice   | model-state       |
| 3   | color              | atomic        | single-choice   | model-state       |
| 4   | position           | atomic        | single-choice   | model-state       |
| 5   | joint-type         | atomic        | single-choice   | model-state       |
| 6   | parent             | atomic        | single-choice   | model-state       |
| 7   | anchor             | atomic        | single-choice   | model-state       |
| 8   | degree             | atomic        | single-choice   | model-state       |
| 9   | recolor            | atomic        | single-choice   | model-state       |
| 10  | add                | atomic        | single-choice   | model-state       |
| 11  | remove             | atomic        | single-choice   | model-state       |
| 12  | replace            | atomic        | single-choice   | model-state       |
| 13  | translate          | atomic        | single-choice   | model-state       |
| 14  | rotate             | atomic        | single-choice   | model-state       |
| 15  | next-module        | atomic        | single-choice   | model-state       |
| 16  | inventory          | atomic        | single-choice   | model-state       |
| 17  | boundary           | atomic        | single-choice   | model-state       |
| 18  | no-op              | atomic        | single-choice   | model-state       |
| 19  | prefix             | metacognitive | single-choice   | support-graph     |
| 20  | access             | metacognitive | multiple-choice | Rapier            |
| 21  | counterfactual     | metacognitive | single-choice   | support-graph     |
| 22  | dynamic            | metacognitive | single-choice   | Rapier            |
| 23  | kinematic          | metacognitive | multiple-choice | model-state       |
| 24  | diagnosis          | metacognitive | multiple-choice | model-state       |
| 25  | information-gain   | metacognitive | multiple-choice | finite-world      |
| 26  | abstention         | metacognitive | single-choice   | finite-world      |
| 27  | posterior          | metacognitive | single-choice   | finite-world      |
| 28  | pareto             | metacognitive | multiple-choice | model-state       |
| 29  | assembly           | procedural    | actions         | state-machine     |
| 30  | disassembly        | procedural    | actions         | state-machine     |
| 31  | service-repair     | procedural    | actions         | state-machine     |
| 32  | compound-edit      | procedural    | actions         | state-machine     |
| 33  | multi-fault        | integrative   | actions         | state-machine     |
| 34  | scheduling         | integrative   | schedule        | resource-schedule |
| 35  | policy             | integrative   | policy          | finite-world      |
| 36  | local-frame        | atomic        | single-choice   | model-state       |
| 37  | projection         | atomic        | single-choice   | model-state       |
| 38  | relative-order     | atomic        | single-choice   | model-state       |
| 39  | joint-axis         | atomic        | single-choice   | model-state       |
| 40  | clearance-margin   | metacognitive | single-choice   | model-state       |
| 41  | load-moment        | metacognitive | single-choice   | model-state       |
| 42  | weighted-posterior | metacognitive | single-choice   | finite-world      |
| 43  | expected-loss      | metacognitive | single-choice   | finite-world      |
| 44  | trace-threshold    | metacognitive | single-choice   | Rapier            |
| 45  | guarded-repair     | procedural    | actions         | state-machine     |
| 46  | rollback           | procedural    | actions         | state-machine     |
| 47  | resource-repair    | integrative   | actions         | state-machine     |
| 48  | budget-policy      | integrative   | policy          | finite-world      |